

11 PFRs

PFR mole balance:

- In terms of number of moles (F): $\frac{dF_A}{dV} = r_A$. In words, the rate of change in the molar flowrate of species A is equal to the rate of generation of A.
 - If species A is a reactant, the rate of generation of A is negative $\frac{1}{2}$ therefore F_A decreases down the reactor volume
 - If species A is a product, the rate of generation of A is positive, so F_A increases down the reactor volume
- In terms of conversion (X): $\frac{dX_A}{dV} = \frac{-r_A}{F_{A0}}$
 - Notice that this balance is written with respect to the rate of consumption of A, because if A is a reactant, then $-r_A$ is positive, so X_A increases as the reactor volume increases
- Applying the PFR mole balances: If you are solving a reactor design problem for a prototypical PFR, you can just write down the PFR mole balance without re-deriving it each time, with the following exceptions.
 - The problem statement asks you to derive the balance.
 - The reactor is not a prototypical PFR.
 - Let's do an example: The reversible reaction $A \rightleftharpoons B$ is carried out isothermally and isobarically in a PFR. The equilibrium constant at the reaction temperature is 5.8. The overall conversion is 0.55. A second PFR that is identical in every way to the first is placed downstream of the first PFR, so that the two PFRs operate in series. Determine the overall conversion for the system. Pure A is fed to the system in both cases.



$$\frac{dX_A}{dV} = \frac{-r_A}{F_{A0}}$$

assume elementary kinetics

$$-r_A = k_A f C_A - k_A r C_B$$

$$-r_c = k_A f \left(C_A - \frac{C_B}{K_c} \right)$$

$$C_A = C_{A0}(1 - X_A)$$

$$C_B = C_{A0} X_A$$

$$\frac{dX_A}{dV} = \frac{k_A f}{F_{A0}} \left[C_{A0}(1 - X_A) - \frac{C_{A0} X_A}{K_c} \right]$$

$$\frac{dX_A}{dV} = \frac{k_A f C_{A0}}{F_{A0}} \left(1 + X_A \left(-1 - \frac{1}{K_c} \right) \right)$$

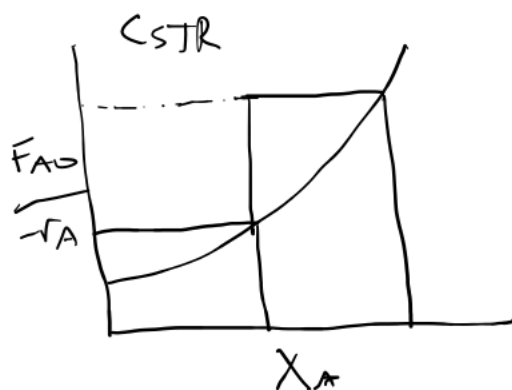
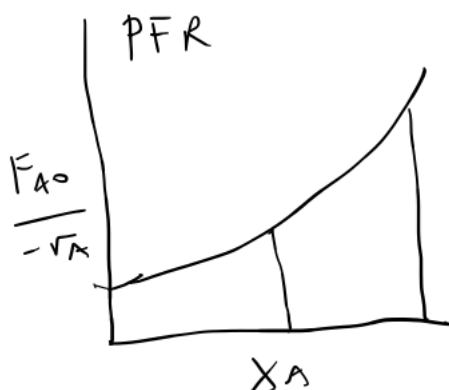
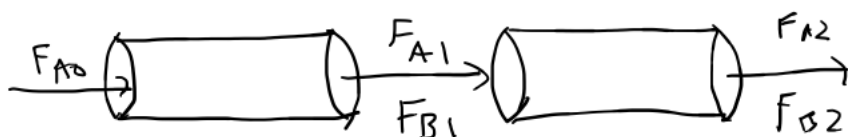
$$\int_0^{0.55} \frac{dX_A}{\left(1 + X_A \left(-1 - \frac{1}{K_C}\right)\right)} = \frac{K_A f C_{A0} V}{F_{A0}}$$

use Appendix A.1 :

$$\int_0^X \frac{dX}{1 + \epsilon X} = \frac{1}{\epsilon} \ln(1 + \epsilon X)$$

here $\epsilon = -1 - \frac{1}{K_C}$, and $X = 0.55$

$$0.883 = \frac{K_A f C_{A0} V}{F_{A0}}$$



$$1.766 = \int_0^X \frac{dX_A}{\left(1 + X_A \left(-1 - \frac{1}{K_C}\right)\right)} = \frac{1}{\epsilon} \ln(1 + \epsilon X)$$

where $\epsilon = -1 - \frac{1}{K_C}$. Solve for X , such that

I get a value of 1.766.

$$x = 0.75$$